

3 Point Attempt and True Shooting Visualizations

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```
library(reticulate)
library(dplyr)
library(ggplot2)
library(tidyr)
```

Data Prep

```
from nba_api.stats.endpoints import leaguedashteamstats
import pandas as pd
import time

def get_season_data(season_year):

    # Advanced stats (for TS%)
    reg = leaguedashteamstats.LeagueDashTeamStats(
        season=season_year,
        season_type_all_star="Regular Season",
        measure_type_detailed_defense="Advanced"
    )
    df_reg = reg.get_data_frames()[0]
    df_reg = df_reg[["TEAM_ID", "TEAM_NAME", "TS_PCT"]]

    time.sleep(1)

    # Base stats (for 3PA = FG3A / FGA)
    reg_base = leaguedashteamstats.LeagueDashTeamStats(
        season=season_year,
        season_type_all_star="Regular Season"
    )
    df_base = reg_base.get_data_frames()[0]
    df_base["THREE_PT_ATTEMPT_RATE"] = df_base["FG3A"] / df_base["FGA"]
    df_base = df_base[["TEAM_ID", "THREE_PT_ATTEMPT_RATE"]]

    time.sleep(1)

    # Playoff win %
    po = leaguedashteamstats.LeagueDashTeamStats(
        season=season_year,
        season_type_all_star="Playoffs"
```

```

)
df_po = po.get_data_frames()[0]
df_po = df_po[["TEAM_ID", "W_PCT"]].rename(columns={"W_PCT": "W_PCT_PO"})

df = pd.merge(df_reg, df_base, on="TEAM_ID")
df = pd.merge(df, df_po, on="TEAM_ID")

return df

years = ["2015-16", "2016-17", "2017-18", "2018-19", "2019-20",
         "2020-21", "2021-22", "2022-23", "2023-24", "2024-25"]

all_season_data = []
for year in years:
    season_df = get_season_data(year)
    season_df["season"] = year
    all_season_data.append(season_df)
    time.sleep(2)

combined_df = pd.concat(all_season_data, ignore_index=True)

```

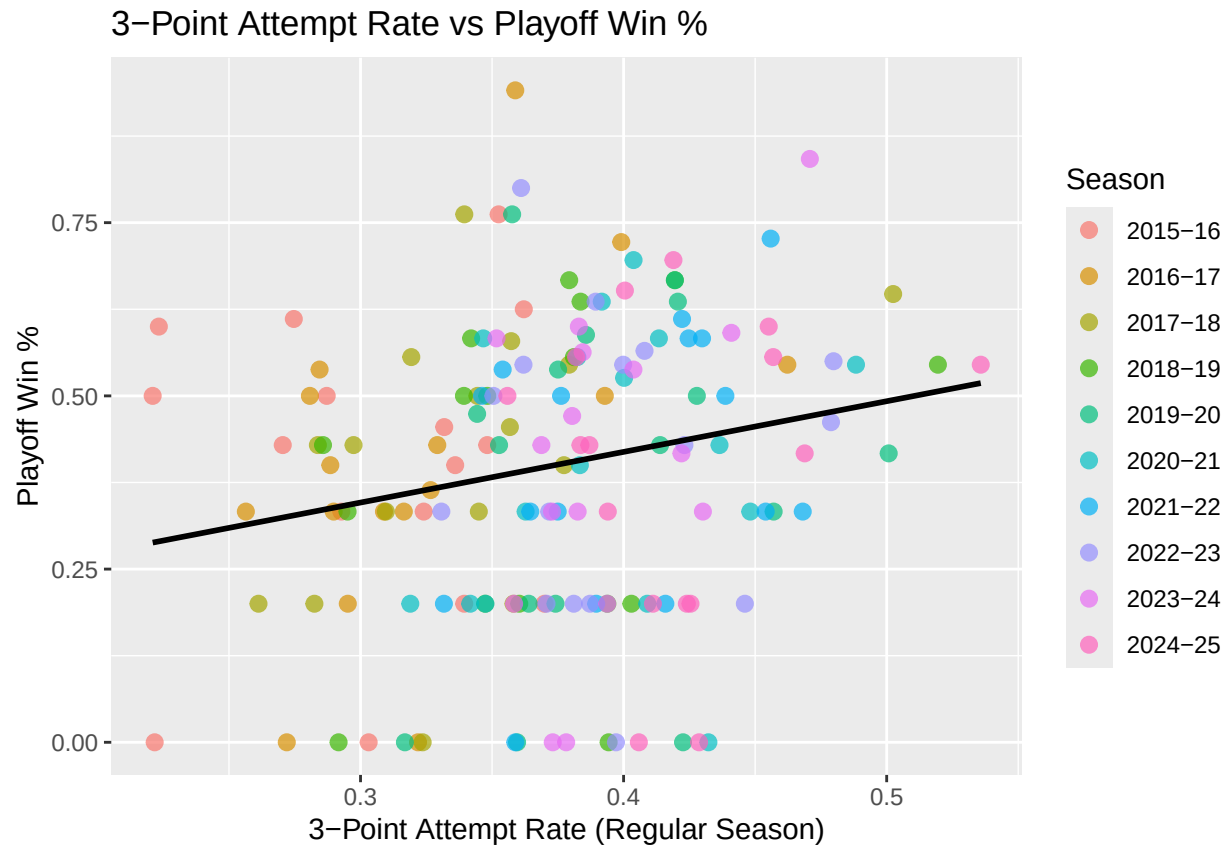
Visualization 1: 3PAR% vs Playoff Win% Scatterplot

```

combined_df <- py$combined_df

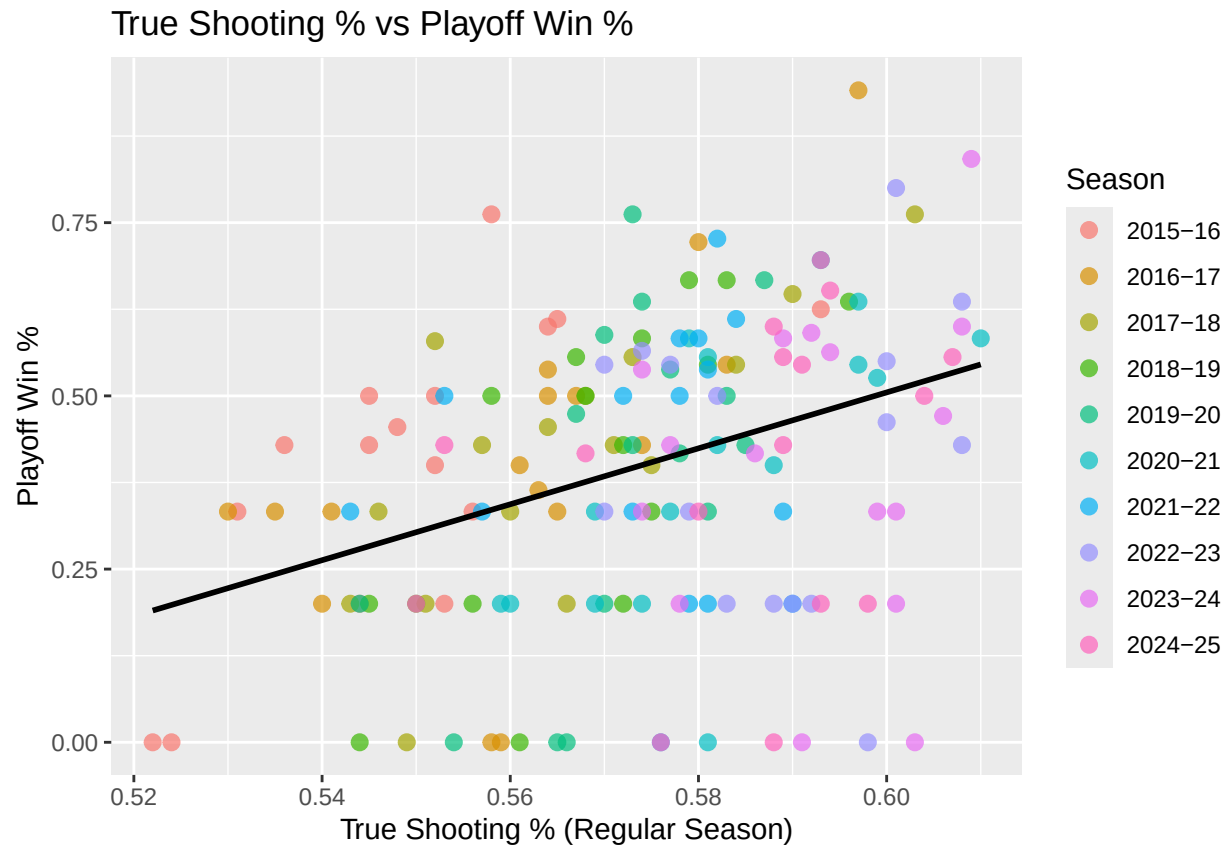
ggplot(combined_df, aes(x = THREE_PT_ATTEMPT_RATE, y = W_PCT_PO, color = season)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(
    title = "3-Point Attempt Rate vs Playoff Win %",
    x = "3-Point Attempt Rate (Regular Season)",
    y = "Playoff Win %",
    color = "Season"
  )

```



Visualization 2: TS% vs Playoff Win% Scatterplot

```
ggplot(combined_df, aes(x = TS_PCT, y = W_PCT_PO, color = season)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  labs(
    title = "True Shooting % vs Playoff Win %",
    x = "True Shooting % (Regular Season)",
    y = "Playoff Win %",
    color = "Season"
  )
```



Visualization 3: Per Season Correlation Barplot

```
cors <- combined_df %>%
  group_by(season) %>%
  summarize(
    cor_3par = cor(THREE_PT_ATTEMPT_RATE, W_PCT_PO),
    cor_ts   = cor(TS_PCT, W_PCT_PO)
  )

cors_long <- cors %>%
  pivot_longer(cols = c(cor_3par, cor_ts),
    names_to = "metric",
    values_to = "correlation") %>%
  mutate(metric = recode(metric,
    cor_3par = "3PT Attempt Rate",
    cor_ts   = "True Shooting %"
  ))

ggplot(cors_long, aes(x = season, y = correlation, fill = metric)) +
  geom_bar(stat = "identity", position = "dodge") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    title = "Per-Season Correlation with Playoff Win %",
  )
```

```

x = "Season",
y = "Correlation",
fill = "Metric"
) +
theme(axis.text.x = element_text(angle = 45, hjust = 1))

```

